

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Library Management System

Requirements Baseline – Version 1.0

Document Information

Item	Details
Project	Library Management System
Organization	
Document	Software Requirements Specification
Version	1.0
Baseline	Requirements Baseline v1.0
Development Approach	Incremental Development with Controlled Change Management
Technology	HTML, CSS, JavaScript, PHP, MySQL
Development Environment	XAMPP
Version Control	Git and GitHub
Status	Proposed for Review
Prepared By	T.Manoja
Date	_____

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for Version 1.0 of the Library Management System.

The document establishes the agreed requirements that will be used as the reference for:

- System analysis
- System design
- Database design
- Software development
- Testing
- Release activities
- Software Change Management

The requirements contained in this document represent the initial proposed scope of the Library Management System.

Once reviewed and approved by the relevant stakeholders, this document will become the Requirements Baseline v1.0.

Any new requirement introduced after the Version 1.0 baseline has been approved shall be handled through the Software Change Management process using a formal Change Request (CR).

1.2 Project Background

The library currently manages information about books, students, and borrowing activities using manual records such as books, registers, or other documents.

Information such as book details, student/member details, issued books, and returned books may be difficult to manage manually.

As the number of books and students increases, managing library records becomes more difficult. The librarian may spend considerable time searching for books, checking student information, recording borrowing transactions, and preparing borrowing reports.

Therefore, the library intends to introduce a web-based Library Management System that provides a centralized method for managing books, students/members, book issuing and returning, and borrowing reports.

The system will allow students to view available books without logging into the system, while librarians will use authenticated login access to manage library operations.

2. Problem Statement

The existing manual library management approach creates several problems:

- Book information is maintained manually.
- Searching for available books can be time-consuming.
- Student/member records may be difficult to manage.
- Recording book borrowing manually can result in errors.
- It can be difficult to identify which student has borrowed a particular book.
- It can be difficult to identify how many students have borrowed a particular book.
- Checking currently issued books requires manual searching.
- Preparing monthly borrowing details requires additional effort.
- Maintaining paper-based records requires physical storage.
- There is no centralized system for managing library information and borrowing records.

The proposed system aims to address these problems by providing a centralized web-based platform for library management.

3. Project Objectives

The main objectives of the Library Management System are:

1. To develop a web-based system for managing library information.
2. To provide librarian authentication.
3. To allow librarians to add, view, update, and delete book records.
4. To allow librarians to add, view, update, and delete and student/member records.
5. To allow librarians to issue books to students.
6. To allow librarians to record returned books.
7. To allow students to view and search for available books without logging in.
8. To maintain accurate borrowing and return records.
9. To provide student-wise, book-wise, and monthly borrowing reports.
10. To reduce manual library record management.
11. To improve the accuracy and consistency of library information.
12. To provide centralized storage of book, student, and borrowing data.
13. To provide a simple and user-friendly web interface.
14. To demonstrate controlled software evolution using Software Change Management practices.

4. Project Scope

4.1 In Scope for Version 1.0

The Version 1.0 system shall provide:

Authentication

- Librarian login

- Librarian logout
- Credential validation
- Secure password storage
- Basic authorization

Student Book Viewing

Students do not require a login.

Students shall be able to:

- View available books
- Search books by id or name, View book details and Check book availability

Book Management

The librarian shall be able to:

- Add books
- View book details list
- Update book information
- Delete book records
- Search books by id or name
- Manage book quantity

Student/Member Management

The librarian shall be able to:

- Add/register students
- View student/member list
- Update student information
- Delete student records

- Search for students by id

Book Issue & Return Management

The librarian shall be able to:

- Select a registered member/student
- Select an available book
- Select the issue date
- Select the due date
- Issue the book
- Save the borrowing transaction list
- Update book availability
- Select the Return action
- View a return confirmation dialog
(Cancel/ Confirm)
- Change the borrowing status to Returned
- Update book availability after return

Reports

The system shall provide:

- Student Borrowing Report – shows the books borrowed by a particular student.
- Book Borrowing Report – shows which students have borrowed a particular book and how many students have borrowed it.
- Overall Borrowing Report – shows all borrowing details, including book and student information.

The report shall include:

- Student ID/Index Number
- Student Name

- Book ID
- Book Name
- Issue Date
- Due Date
- Return Date
- Borrowing Status

Technical Scope.

- HTML/CSS/JavaScript frontend
- PHP backend
- MySQL database
- XAMPP development environment
- Git/GitHub version control

4.2 Out of Scope for Version 1.0

The following functionality is intentionally excluded from Version 1.0:

- Student login
- Online book reservation
- Online book renewal by students
- Online fine payment
- Email/SMS notifications for Overdue
- Barcode/QR-code scanning
- E-book management
- Mobile application

These features may be introduced in future releases through approved Change Requests.

5. Stakeholders

5.1 Stakeholder Register

Stakeholder	Role	Interest / Responsibility
Librarian	System User	Manage books, students, issuing, returning and reports
Student	End User	View and search available books
School/Institute Management	Management	Monitor library operations and system development
IT Administrator	Technical Stakeholder	Maintain and support the system
Project Team	Development Team	Analyze, design, develop and test the system
Lecturer/Supervisor	Academic Supervisor	Monitor project progress and requirements

6. User Roles

6.1 Librarian

The librarian is the main authorized user of the Library Management System.

The librarian shall be able to:

- Log in.
- Log out.
- Access the librarian dashboard.
- Add books.

- View book list.
- Update book information.
- Delete book records.
- Search books.
- Add/register students.
- View student records list.
- Update student information.
- Delete student records.
- Search students.
- Issue books.
- Record returned books.
- View currently issued books list.
- Return books.
- Confirm book returns..
- Generate student borrowing reports.
- Generate book borrowing reports.
- Generate Over all borrowing reports.

Only an authenticated librarian shall be permitted to access these management functions.

6.2 Student

The student is a public/end user of the book-viewing part of the system.

Student login is not required.

The student shall be able to:

- Open the library system.
- View available books.
- Search for books.
- View book details.
- Check whether a book is available.

Students shall not be permitted to:

- Add books.
- Update books.
- Delete books.
- Add students.
- Update student records.
- Delete student records.
- Issue books through the system.
- View other students' borrowing information.
- Access librarian reports.
- Access the librarian dashboard.

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7. Functional Requirements

Functional requirements describe what the Library Management System shall do.

7.1 Authentication Requirements

FR-01 – Librarian Login

The system shall allow an authorized librarian to log in using a valid username and password.

FR-02 – Login Validation

The system shall validate the username and password provided by the librarian.

FR-03 – Invalid Login Handling

The system shall display an appropriate error message when invalid librarian login credentials are provided.

FR-04 – Librarian Logout

The system shall allow an authenticated librarian to log out of the system.

FR-05 – Protected Access

The system shall prevent unauthenticated users from accessing librarian management functions.

7.2 Book Viewing Requirements

FR-06 – View Available Books

The system shall allow students to view books that are currently available in the library without logging in.

FR-07 – Search Books

The system shall allow students and librarians to search for books using relevant information such as Book ID.

FR-08 – View Book lists

The system allow librarians to view the list.

FR-09 – Display Book Availability

The system shall display the current availability status of a book

7.3 Book Management Requirements

FR-10 – Add Book

The system shall allow an authenticated librarian to add a new book record.

FR-11 – Record Book Information

The system should store relevant book information such as Book ID, title, author, ISBN, category, and quantity.

FR-12 – View Book List

The system shall allow the librarian to view all registered books.

FR-13 – Update Book

The system shall allow the librarian to update existing book information.

FR-14 – Delete Book

The system shall allow the librarian to delete a book record when appropriate.

FR-15 – Search Book

The system shall allow the librarian to search for books

7.4 Student/Member Management Requirements

FR-17 – Add Student/Member

The system shall allow an authenticated librarian to register a new student/member.

FR-18 – View Student List

The system shall allow the librarian to view the list of registered students/members.

FR-19– View Student Details

The system shall allow the librarian to view the details of a selected student/member.

FR-20 – Update Student

The system shall allow the librarian to update student/member information.

FR-21 – Delete Student

The system shall allow the librarian to delete a student/member record when appropriate.

FR-22 – Search Student

The system allows the librarian to search for students using information such as Index Number.

7.5 Issue & Return Management Requirements

FR-22 – Issue Book

The system shall allow the librarian to issue an available book to a registered member.

FR-23 – Check Book Availability

The system should check whether the selected book is available before issuing it.

FR-24 – Record Issue Details

The system shall record the member, book, issue date, and due date.

FR-25 – View Issued Books list

The system shall allow the librarian to view issued book records.

FR-26 – Return Action

The system shall provide a Return action for currently borrowed books.

FR-27 – Return Confirmation

When the librarian clicks Return, the system shall display a confirmation

FR-28 – Confirm Return

When the librarian selects Confirm Return, the system shall:

- Change the status to Returned.

- Update book availability.
- Save the updated borrowing record.

FR-29 – Cancel Return

When the librarian selects Cancel, the system shall close the confirmation dialog without changing the borrowing record.

FR-30 – Prevent Issue of Unavailable Book

The system shall prevent a book from being issued when it is not available

7.6 Reporting Requirements

FR-31 – Student Borrowing Report

The system shall allow the librarian to view the borrowing details of a particular student/member.

FR-32 – Particular Book Borrowing Report

The system shall allow the librarian to see which students have borrowed a particular book.

FR-33 – Borrowing Report

The system shall allow the librarian view all borrowing details

The report shall include:

- Student/Member
- Book
- Issue Date
- Due Date

8. Non-Functional Requirements

8.1 Security

NFR-01 – Password Security

The system shall store librarian passwords using secure password hashing.

NFR-02 – Authentication

Protected librarian functions require successful authentication.

NFR-03 – Authorization

The system shall restrict book management, student management, issue/return management, and reporting functions to authorized librarians.

8.2 Performance

NFR-04 – Response Time

Under normal operating conditions, commonly used pages and database operations should respond within approximately three seconds.

NFR-05 – Database Performance

The database shall be designed to support efficient retrieval, insertion, updating, and deletion of library records.

6. User Roles

NFR-06 – User-Friendly Interface

The system should provide a simple and easy-to-understand interface.

NFR-07 – Validation Messages

The system should provide clear messages when user input is invalid, or an operation cannot be completed.

8.4 Reliability

NFR-08 – Data Integrity

The system shall maintain the integrity and consistency of books, students, and borrowing information.

NFR-09 – Error Handling

The system shall handle expected application errors without causing the application to terminate unexpectedly.

8.5 Availability

NFR-12 – Availability

The system should be available during the library's normal operating hours when deployed in the target environment.

8.6 Maintainability

NFR-13 – Maintainable Structure

The system should use a structured and modular PHP application design that allows future changes to be implemented with minimal unnecessary impact on existing functionality.

8.7 Portability

NFR-14 – Browser Compatibility

The web application should operate correctly on commonly used modern web browsers.

8.8 Responsiveness

NFR-15 – Responsive Interface

The user interface should adapt to common desktop, tablet, and mobile screen sizes.

9. User Stories

US-01 – Librarian Login

As a librarian,
I want to log in using my credentials,
so that I can securely access the library management functions.

US-02 – View Available Books

As a student,
I want to view available books without logging in,
so that I can find books that are available in the library.

US-03 – Search Books

As a student,
I want to search for books,
so that I can quickly find a required book.

US-04 – Add Book

As a librarian,
I want to add a new book,
so that the book information is stored in the library system.

US-05 – View Books

As a librarian,
I want to view the book list,
so that I can manage the library's books.

US-06 – Update Book

As a librarian,
I want to update book information,
so that the records remain accurate.

US-07 – Delete Book

As a librarian,
I want to delete a book record,
so that unnecessary book records can be removed.

US-08 – Register Student

As a librarian,
I want to register as a student/member,
so that the student can borrow books from the library.

US-09 – Search Student

As a librarian,
I want to search for a student,
so that I can quickly find the student's library record.

US-10 – Issue Book

As a librarian,
I want to issue an available book to a student,
so that the borrowing transaction is recorded.

US-11 – Return Book

As a librarian,
I want to record a returned book,
so that the book becomes available again and the borrowing record is updated.

US-12 – View Student Borrowing History

As a librarian,
I want to view the books borrowed by a particular student,
so that I can check the student's borrowing history.

US-13 – View Book Borrowing History

As a librarian,
I want to view which students have borrowed a particular book,
so that I can understand the borrowing history of that book.

US-14 – View Borrowing Report

As a librarian,
I want to select a month and view all borrowing transactions during that month,
so that I can monitor the library's monthly borrowing activity.

US-15 – Librarian Logout

As a librarian,
I want to log out of the system,
so that my account remains protected after I finish using the system.

10. Acceptance Criteria

AC-01 – Librarian Login

Given a registered librarian is on the login page,
When the librarian enters valid credentials and selects Login,
Then the system shall authenticate the librarian and display the Librarian Dashboard.
If invalid credentials are entered, an appropriate error message shall be displayed.

AC-02 – View Available Books

Given a student opens the library system,
When the student selects the available books section,
Then the system should display books that are currently available.
The student shall not be required to log in.

AC-03 – Search Books

Given a user is viewing the book list,

When the user enters a valid search value,

Then the system shall display matching book records.

If no matching book exists, the system shall display an appropriate message.

AC-04 – Add Book

Given the librarian is authenticated,

When the librarian enters valid book information and selects Save,

Then:

- The book record shall be saved.
 - A unique Book ID shall be recorded.
 - A success message shall be displayed.
 - The new book shall appear in the book list.
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AC-05 – Update Book

Given the librarian is authenticated,

When the librarian modifies valid book information and selects Update,

Then:

- The existing book record shall be updated.
 - The updated information shall be stored in the database.
 - A success message shall be displayed.
 - The updated information shall be visible when the book is viewed again.
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AC-06 – Delete Book

Given the librarian is authenticated,

When the librarian selects a book and confirms deletion,

Then:

- The book record shall be removed.
 - The book shall no longer appear in the book list.
 - A confirmation message shall be displayed.
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AC-07 – Add Student

Given the librarian is authenticated,

When the librarian enters valid student information and selects Save,

Then:

- The student record shall be saved.
 - The student shall appear on the student list.
 - A success message shall be displayed.
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AC-08 – Search Student

Given the librarian is authenticated,

When the librarian enters a valid Index Number.

Then the system shall display matching student records.

If no matching student exists, the system shall display an appropriate message.

AC-09 – Issue Book

Given the librarian is authenticated,

When the librarian selects a registered student and an available book and enters valid issue information,

Then:

- The borrowing record shall be saved.
- The issue date shall be recorded.
- The due date shall be recorded.
- The book availability shall be updated.
- A successful issue message shall be displayed.

If the selected book is unavailable, the system shall prevent the issue transaction.

AC-10 – Return Book

Given a book has previously been issued,

When the librarian records the return,

Then:

- The return date shall be recorded.
 - The borrowing status shall be updated to Returned.
 - The book availability shall be updated.
 - A successful return message shall be displayed.
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AC-08 – Confirm Return

Given the Return Confirmation dialog is displayed,

When the librarian clicks Confirm Return,

Then the system shall:

- Change the status to Returned.
- Update book availability.

- Save the changes.
 - Display a success message.
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AC-09 – Cancel Return

Given the Return Confirmation dialog is displayed,

When the librarian clicks Cancel,

Then the dialog shall close and the borrowing record shall remain unchanged.

AC-11 – Student Borrowing Report

Given the librarian is authenticated,

When the librarian selects a particular student,

Then the system shall display the books borrowed by that student together with the relevant issues, due, return, and status information.

AC-12 – Particular Book Borrowing Report

Given the librarian is authenticated,

When the librarian selects a particular book,

Then the system shall display the students who have borrowed that book and the relevant borrowing details.

AC-13 – Borrowing Report

Given the librarian is authenticated,

When the librarian selects a particular month,

Then the system shall display all borrowing transactions recorded during that month.

The report shall include both student and book details together with issue, due, return, and status information.

AC-16 – Logout

Given a librarian is authenticated,

When the librarian selects Logout,

Then the system shall terminate the authenticated session and return the librarian to the login page.

11. MoSCoW Prioritization

The Version 1.0 requirements are prioritized using the MoSCoW method.

Priority	Requirements
Must Have	Librarian Login, Logout, Book Management, Student/Member Management, Book Issue, Book Return, Book Search, Available Book Viewing
Should Have	Student Search, Book Search by Category/Author, Currently Issued Books Report
Could Have	Student Borrowing Report, Particular Book Borrowing Report, Borrowing Report, Basic Dashboard Statistics
Won't Have in v1.0	Student Login, Online Reservation, Online Renewal, Online Fine Payment, Notifications, Barcode/QR Scanning, Mobile Application, E-book Management

"Won't Have" means that the functionality is excluded from Version 1.0. It does not mean that the functionality can never be developed.

12. Product Backlog

ID	Product Backlog Item	Priority	Related Requirements	Initial Status
US-01	Librarian Login	Must	FR-01, FR-02, FR-03	Planned
US-15	Librarian Logout	Must	FR-04, FR-05	Planned
US-02	View Available Books	Must	FR-06, FR-09	Planned
US-03	Search Books	Must	FR-07, FR-16	Planned
US-04	Add Book	Must	FR-10, FR-11	Planned
US-05	View Books	Must	FR-12, FR-13	Planned
US-06	Update Book	Must	FR-14	Planned
US-07	Delete Book	Must	FR-15	Planned
US-08	Register Student	Must	FR-18	Planned
US-09	Search Student	Should	FR-23	Planned
US-10	Issue Book	Must	FR-24–FR-28	Planned
US-11	Return Book	Must	FR-29–FR-34	Planned

US-12	Student Borrowing Report	Could	FR-35	Planned
US-13	Particular Book Borrowing Report	Could	FR-36	Planned
US-14	Borrowing Report	Could	FR-37	Planned
US-15	Currently Issued Books	Should	FR-38	Planned

13. Use Cases

13.1 Use Case List

Use Case ID	Use Case	Primary Actor
UC-01	Librarian Login	Librarian
UC-02	View Available Books	Student
UC-03	Search Books	Student / Librarian
UC-04	Add Book	Librarian
UC-05	View Books	Librarian
UC-06	Update Book	Librarian
UC-07	Delete Book	Librarian
UC-08	Add Student/Member	Librarian
UC-09	View Students	Librarian
UC-10	Search Student	Librarian
UC-11	Update Student	Librarian
UC-12	Delete Student	Librarian

UC-13	Issue Book	Librarian
UC-14	Return Book	Librarian
UC-15	Student Borrowing Report	Librarian
UC-16	Particular Book Borrowing Report	Librarian
UC-17	Monthly Borrowing Report	Librarian
UC-18	Currently Issued Books Report	Librarian
UC-20	Librarian Logout	Librarian

13.2 Use Case Example – Issue Book

Field	Description
Use Case ID	UC-13
Name	Issue Book
Actor	Librarian
Preconditions	Librarian must be authenticated and the student must be registered
Trigger	Librarian selects Issue Book
Main Flow	Select student → Select the availability book → Enter issue date → Enter due date → Validate information → Save borrowing record → Update book availability → Display confirmation
Alternative Flow	Selected books are unavailable or student/book information is invalid
Postcondition	Borrowing record is stored successfully and book availability is updated

14. Business Rules

The following business rules apply to Version 1.0.

BR-01

Only authenticated librarians may access protected library management functions.

BR-02

Only authorized librarians may add, update, or delete book records.

BR-03

Only authorized librarians may add, update, or delete member records.

BR-04

Each book shall have a unique Book ID.

BR-05

Each member/student shall have a unique Index Number or Member ID.

BR-06

A book shall only be issued when it is available.

BR-07

A borrowing transaction shall be associated with one member and one book.

BR-08

The issue date shall be recorded when a book is issued.

BR-09

A due date shall be recorded for each borrowing transaction.

BR-10

When a book is returned, the librarian must confirm the return before the transaction is updated.

BR-11

The system shall record the actual return date after the librarian confirms the return.

BR-12

A returned book shall be updated as available.

BR-13

A completed borrowing transaction shall have a Returned status.

BR-14

Students shall not be required to log in to view available books.

BR-15

Students shall not be permitted to access librarian management functions.

BR-16

Students shall not be permitted to view other students' borrowing information.

BR-17

The Monthly Borrowing Report shall display all borrowing transactions recorded during the selected month.

BR-18

An overdue book shall be identified when its due date has passed and it has not been returned.

BR-19

Required books, members, and borrowing information must be provided before a record can be saved.

BR-20

Librarian passwords shall not be stored as plain text.

The original SRS similarly uses numbered business rules to define system restrictions and data rules

15. Data Requirements

Version 1.0 requires the system to maintain the following core information.

15.1 Librarian Information

- Librarian ID
- Username
- Password

15.2 Student/Member Information

- Student ID
- Index Number
- Student Name
- Grade
- Contact Number
- Registration Date

15.3 Book Information

- Book ID
- Book Title
- Author
- Category
- Quantit

- Book Status

15.4 Borrowing Information

- Issue ID
- Student ID
- Book ID
- Issue Date
- Due Date
- Return Date
- Borrowing Status

The database structures, relationships, normalization, constraints, and SQL implementation will be documented separately during the Database Design phase.

16. System Constraints

The Version 1.0 project has the following constraints:

1. The application will be developed as a web-based system.
2. HTML will be used to structure the web pages.
3. CSS will be used for styling and responsive interface design.
4. JavaScript will be used for client-side interactions and validation.
5. PHP will be used for server-side application development.
6. MySQL will be used as the database management system.
7. XAMPP will be used as the local development environment.
8. Git and GitHub may be used for version control.
9. Version 1.0 will focus on essential library management functionality.
10. New requirements after baseline approval must follow the Change Request process.

11. The system will require a web browser to access the application.
 12. The system will depend on the PHP and MySQL services provided through XAMPP during local development.
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17. Assumptions

The following assumptions have been made:

1. The library will provide authorized librarian users to operate and test the system.
 2. Librarians will receive valid login credentials.
 3. The library will provide accurate book information.
 4. The library will provide accurate member/student information.
 5. Students will have access to a modern web browser.\
 6. Students will not require login credentials to view available books.
 1. Book information entered into the system will be provided by authorized library personnel.
 2. Student/member information entered into the system will be provided by authorized library personnel.
 3. The library will define the borrowing period and due-date rules.
 4. The library will provide accurate information about books and students.
 5. XAMPP will be available during local development and testing.
 6. MySQL will be used to store library system data.
 7. Future functionality can be introduced through controlled software changes.
 8. Requirements may evolve after Version 1.0 and will therefore be managed through the Software Change Management process.
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18. Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) provides traceability between functional requirements, user stories, acceptance criteria, and future testing activities.

Requirement	User Story	Acceptance Criteria	Test Case
FR-01 Login	US-01	AC-01	TC-01
FR-04 Logout	US-17	AC-16	TC-02
FR-06 View Available Books	US-02	AC-02	TC-03
FR-07 Search Books	US-03	AC-03	TC-04
FR-10 Add Book	US-04	AC-04	TC-05
FR-12 View Books	US-05	AC-04	TC-06
FR-14 Update Book	US-06	AC-05	TC-07
FR-15 Delete Book	US-07	AC-06	TC-08
FR-18 Add Student	US-08	AC-07	TC-09
FR-23 Search Student	US-09	AC-08	TC-10
FR-24 Issue Book	US-10	AC-09	TC-11
FR-31 Record Book Return	US-11	AC-10	TC-12
FR-35 Student Borrowing Report	US-12	AC-11	TC-13
FR-36 Particular Book Report	US-13	AC-12	TC-14
FR-37 Borrowing Report	US-14	AC-13	TC-15
FR-38 Issued Books Report	US-15	AC-14	TC-16

The test case identifiers will be formally defined and expanded during the Test Planning and Testing sessions.

19. Requirements Review

Before establishing the baseline, the requirements shall be reviewed with the relevant stakeholders.

Review Question 1

Can students log in to the system?

Decision: No.

Students do not require login. They can view and search available books without authentication.

Review Question 2

Can students add, update, or delete books?

Decision: No.

Only authorized librarians can manage book records.

Review Question 3

Can students add, update, or delete student records?

Decision: No.

Only authorized librarians can manage student/member records.

Review Question 4

Can students directly issue books through the system?

Decision: No.

The librarian records book issue transactions.

Review Question 5

Can a librarian issue an unavailable book?

Decision: No.

The system shall check book availability before completing an issue transaction.

Review Question 6

Can students view other students' borrowing information?

Decision: No.

Borrowing information is restricted to authorized librarians.

Review Question 7

Are borrowing reports included in Version 1.0?

Decision: Yes.

Version 1.0 includes student borrowing reports, particular book borrowing reports, monthly borrowing reports, currently issued book reports, and overdue book reports.

Review Question 8

What does the Monthly Borrowing Report contain?

Decision:

The Monthly Borrowing Report shall display all borrowing transactions recorded during the selected month, including student details, book details, issue date, due date, return date, and borrowing status.

Review Question 9

Can a student edit library records?

Decision: No.

The student role is limited to viewing and searching available book information.

20. Version 1.0 Requirements Baseline

After stakeholder review and approval, the requirements contained in this document shall become the Requirements Baseline v1.0.

The baseline establishes the official reference point for Version 1.0 development.

After baselining:

- New requirements shall not be added directly to the Version 1.0 scope.
- Any new requirement must be submitted through the Software Change Management process.
- Approved changes shall be assessed for their impact on requirements, design, database, implementation, testing, and documentation.
- The relevant baselines and information shall be updated when an approved change is implemented.

The general change process shall be:

New Requirement → Change Request → Impact Analysis → Change Approval → Implementation → Testing → Code Review → Merge → Documentation Update → New Version / Release

21. Configuration and Version Control

The approved requirements document shall be maintained under Git version control.

21.1 Baseline Information

Item	Value
Baseline Name	Requirements Baseline v1.0
Version	1.0
Status	Proposed for Review / Approved after Sign-Off
Git Tag	requirements-v1.0
Baseline Date	_____
Approved By	_____

The baseline tag requirements-v1.0 will provide a fixed reference point for future change analysis.

22. Future Change Management

Once the Version 1.0 requirements have been baselined, future requirements may be introduced by the library through the formal change management process.

Example Future Request

"The library now requires students to reserve books online."

This requirement shall not be automatically added to Version 1.0.

Instead, the request shall follow the formal change process:

Customer Request → CR-001 → Impact Analysis → CCB / Project Approval → Requirements Update → Design Update → Database Update → Backend Update → Frontend Update → Testing → Release v1.1

This mechanism will allow the project to demonstrate the complete Software Change Management lifecycle.

23. Version History

Version	Date	Description	Author	Status
0.1	_____	Initial requirements draft	_____	Draft
0.2	_____	Requirements reviewed and revised	_____	Review
1.0	_____	Requirements baseline established	_____	Approved

24. Approval

By approving this document, the relevant stakeholders confirm that the requirements defined for Version 1.0 represent the agreed initial scope of the Library Management System.

Role	Name	Signature	Date
Client / Library Representative	_____ —	_____ —	_____ —
Project Team Representative	_____ —	_____ —	_____ —
Lecturer / Supervisor	_____ —	_____ —	_____ —

Appendix A – Version 1.0 Scope Summary

Included

- Librarian Authentication
- Librarian Login

- Librarian Logout
- Available Book Viewing
- Book Searching
- Book Management
- Book Registration
- Book Updating
- Book Deletion
- Student/Member Registration
- Student/Member Management
- Student Searching
- Book Issue
- Book Return
- Book Availability Management
- Student Borrowing Report
- Particular Book Borrowing Report
- Monthly Borrowing Report
- Currently Issued Books Report
- Overdue Books Report
- Basic Security
- Basic Role-Based Access
- Responsive Web Interface
- HTML/CSS/JavaScript Frontend
- PHP Backend
- MySQL Database

- XAMPP Development Environment

Not Included

- Student Login
- Online Book Reservation
- Online Book Renewal
- Online Fine Payment
- Email/SMS Notifications
- Barcode/QR-Code Scanning
- Mobile Application
- E-book Management
- Multiple Library Branch Management
- Advanced Analytics
- Advanced Audit Logs
- Advanced Role-Based Access Control
- Cloud/Production Deployment

Appendix B – Initial Technology Environment

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	PHP
Database	MySQL
Development Environment	XAMPP
Version Control	Git
Repository	GitHub

Appendix C – Project Lifecycle

The initial project shall follow the following lifecycle:

Project Scenario → Requirements Engineering → Requirements Baseline → System Analysis & Design → Design Baseline → Database Design → Database Baseline → PHP Backend Development → Frontend Development → Integration → Testing → Version 1.0 Release

After the first release:

Change Request → Impact Analysis → Approved Change → Implementation → Testing → New Release

Appendix D – Baseline Statement

Requirements Baseline v1.0

The requirements contained in this document constitute the proposed Version 1.0 requirements baseline for the Library Management System.

Any modification to these requirements after baseline approval must be controlled through the Software Change Management process.

Baseline Information

Item	Details
Baseline Tag	requirements-v1.0
Baseline Date	_____
Approved By	_____
Approval Signature	_____

Appendix E – Project Document Sequence

The Requirements Baseline is positioned within the overall practical project as follows:

01 – Initial Client Scenario → 02 – Initial Project Proposal → 03 – Project Plan / Practical Roadmap → 04 – SRS + Requirements Baseline v1.0 → Session 3 – SCM Setup → Session 4 – System Design → Session 5 – Database Design → Development → Testing → Version 1.0 Release → CR-001 → Impact Analysis → Version 1.1

Final Baseline Note

- Version 1.0 focuses on librarian login, book management, student/member management, issue and return management, and borrowing reports.
- Students can view and search for available books without logging in.
- Features such as student login, online reservation, notifications, mobile application, and barcode/QR scanning are excluded from Version 1.0.
- Future features can be added through formal Change Requests.
- Overdue books.

The Monthly Borrowing Report specifically provides a complete record of borrowing activity for the selected month, containing both student and book information together with issue, due, return, and status details.

Any modification to these requirements after baseline approval shall be managed through the formal Software Change Management process.